WORKED SOLUTION

STEP I 2005, Question 14

Probability and Statistics • Mixed Random Variables

STEP I 2005, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q14

The key idea

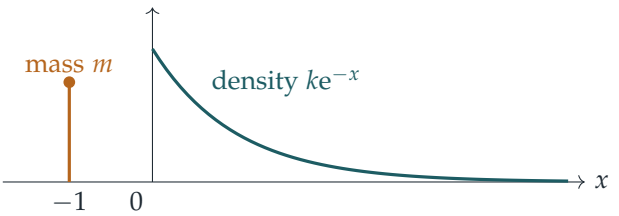
X is a *mixed* random variable: a lump of probability sitting at the single point -1, together with a continuous spread over $[0, \infty)$. The way to handle it is to split every calculation into the two pieces: a discrete term from $X = -1$ plus an integral against the density on $[0, \infty)$. Once we notice that the continuous piece is just a scaled exponential distribution, most of the integrals are ones we already know.

Solution

Differentiating the given cumulative probability, the continuous part of X has density

$$f(x) = \frac{\mathrm{d}}{\mathrm{d}x} k(1 - \mathrm{e}^{-x}) = k\mathrm{e}^{-x}, \qquad x \geq 0.$$

So X looks like this: a point mass m at -1, and a curve $k\mathrm{e}^{-x}$ on $[0, \infty)$.



Part (i)

The total probability is 1. Letting $x \to \infty$, $\mathrm{P}(X \geq 0) = k(1 - 0) = k$, so

$$m + k = 1, \qquad \text{that is,} \qquad k = 1 - m.$$

(Since $m \geq 0$ is a probability and $m < \frac{1}{2}$, we have $k > \frac{1}{2} > 0$, so the density is genuinely positive.)

Part (ii)

Splitting into the two pieces,

$$\begin{aligned} \mathrm{E}(X) &= (-1)\cdot m + \int_0^\infty x\,(1-m)\mathrm{e}^{-x}\,\mathrm{d}x \\ &= -m + (1-m)\cdot 1 = 1-2m, \end{aligned}$$

using $\int_0^\infty x\mathrm{e}^{-x}\,\mathrm{d}x = 1$ (integrate by parts, or recognise the mean of an exponential distribution with parameter 1). This is the required result.

Part (iii)

Variance. In the same way, using $\int_0^\infty x^2\mathrm{e}^{-x}\,\mathrm{d}x = 2$ (integrate by parts twice),

$$\mathrm{E}(X^2) = (-1)^2\cdot m + (1-m)\cdot 2 = 2-m.$$

Hence

$$\begin{aligned} \mathrm{Var}(X) &= \mathrm{E}(X^2) - (\mathrm{E}(X))^2 = 2 - m - (1-2m)^2 \\ &= 1 + 3m - 4m^2 = (1-m)(1+4m). \end{aligned}$$

Ask yourself Could the median be -1, or lie somewhere between -1 and 0?

Median. No: X never takes values in $(-1,0)$, and $\mathrm{P}(X \le -1) = m < \frac{1}{2}$, so the point mass alone is not enough to reach half the probability. This is exactly where the condition $m < \frac{1}{2}$ is used. So the median M satisfies $M \ge 0$ and $\mathrm{P}(X \le M) = \frac{1}{2}$:

$$\begin{aligned} m + (1-m)(1-\mathrm{e}^{-M}) &= \tfrac{1}{2} \\ 1 - (1-m)\mathrm{e}^{-M} &= \tfrac{1}{2} \\ \mathrm{e}^{-M} &= \frac{1}{2(1-m)}. \end{aligned}$$

Therefore

$$M = \ln\left(2(1-m)\right) = \ln(2-2m).$$

As a consistency check, $0 \le m < \frac{1}{2}$ gives $1 < 2-2m \le 2$, so $M > 0$, as it must be. Because the distribution is continuous on $[0,\infty)$, this M also has $\mathrm{P}(X \ge M) = \frac{1}{2}$, so the median is unambiguous.

Part (iv)

At $X = -1$ we have $|X|^{1/2} = 1$, and for $X \ge 0$ we have $|X|^{1/2} = \sqrt{X}$. So

$$\mathrm{E}(|X|^{1/2}) = 1\cdot m + (1-m)\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x.$$

Ask yourself How do we turn $\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x$ into the integral we were given?

The given integral has e^{-y^2}, which suggests putting $x = y^2$ (with $y \geq 0$), so $\sqrt{x} = y$ and $\mathrm{d}x = 2y\,\mathrm{d}y$; the limits 0 and ∞ are unchanged:

$$\int_0^\infty \sqrt{x}\,\mathrm{e}^{-x}\,\mathrm{d}x = \int_0^\infty y\,\mathrm{e}^{-y^2}\,2y\,\mathrm{d}y = 2\int_0^\infty y^2\mathrm{e}^{-y^2}\,\mathrm{d}y = 2\cdot\tfrac{1}{4}\sqrt{\pi} = \tfrac{1}{2}\sqrt{\pi}.$$

Hence

$$\mathrm{E}(|X|^{1/2}) = m + \tfrac{1}{2}\sqrt{\pi}\,(1-m).$$

Checks

- With $m = 0$, X is exactly exponential with parameter 1: our formulas give mean 1, variance 1, median $\ln 2$ and $\mathrm{E}(\sqrt{X}) = \frac{1}{2}\sqrt{\pi} = \Gamma(\frac{3}{2})$, all the standard values.
- As $m \to \frac{1}{2}$ the median tends to $\ln 1 = 0$, which makes sense: half the probability is at -1, so the median slides down to the bottom of the continuous part.
- A simulation with $m = 0.3$ and 4×10^5 samples gave mean 0.402 (formula 0.4), variance 1.551 (formula 1.54), median 0.336 (formula $\ln 1.4 \approx 0.336$) and $\mathrm{E}(|X|^{1/2}) \approx 0.922$ (formula 0.920).

Common slips

- Forgetting the point mass in $\mathrm{E}(X^2)$ or $\mathrm{E}(|X|^{1/2})$: it contributes $+m$, not $-m$, because $(-1)^2 = |-1|^{1/2} = 1$.
- Solving $(1-m)(1-\mathrm{e}^{-M}) = \frac{1}{2}$ for the median, which ignores the probability m already accumulated at -1.
- Changing variable without converting $\mathrm{d}x$ to $2y\,\mathrm{d}y$, which gives $\frac{1}{4}\sqrt{\pi}$ instead of $\frac{1}{2}\sqrt{\pi}$.

Beyond STEP

Distributions that mix point masses and densities are common in practice (an insurance claim is zero with some probability, otherwise continuous). At university they are handled uniformly by writing expectations as integrals against the distribution function, $\mathrm{E}(g(X)) = \int g(x)\,\mathrm{d}F(x)$, which covers discrete, continuous and mixed cases at once. The integral in part (iv) is the Gamma function value $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$. Grimmett and Welsh, *Probability: An Introduction*, is a good next step.

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